

Resilient Healthcare

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead / Applicant]
Dr Saba Hinrichs-Krapels	TU Delft (TUD)	Technology, Policy and Management (TPM), Multi-Actor Systems	S.Hinrichs@tudelft.nl	Tenure-track/Delft Technology Fellow	–	Main lead
Professor Robert Baatenburg de Jong	Erasmus MC (EMC)	Otorhinolaryngology and Head & Neck Surgery	r.j.baatenburgdejong@erasmusmc.nl	Tenured	Mix	Lead
Professor Hester Lingsma	Erasmus MC (EMC)	Public Health	h.lingsma@erasmusmc.nl	Tenured	Researcher	Applicant / Lead (takes over when Prof Baatenburg de Jong retires)
Professor Roland Bal	Erasmus University Rotterdam (EUR)	ESPHM, Healthcare governance group	r.bal@eshpm.eur.nl	Tenured	–	Lead
Dr Irene Grossmann	TU Delft (TUD)	Technology, Policy and Management (TPM), Centre for Safety in Healthcare	i.grossmann@tudelft.nl	tenure track	–	Applicant
Professor Tina Comes	TU Delft (TUD)	TPM, Engineering Systems & Services	t.comes@tudelft.nl	tenured	–	Applicant
Dr Neil Yorke-Smith	TU Delft (TUD)	Software Technology	n.yorke-smith@tudelft.nl	tenured	–	Applicant
Dr Theresia van Essen	TU Delft (TUD)	Applied Mathematics	j.t.vanessen@tudelft.nl	tenured	–	Applicant
Dr. Kateřina Staňková	TU Delft (TUD)	TPM, department Engineering Systems and Services	K.Stankova@tudelft.nl	Other (Delft Technology Fellowship)	–	Applicant
Professor Jop Groeneweg	TU Delft (TUD)	TPM	j.groeneweg-1@tudelft.nl	tenured	–	Applicant
Professor Alexander Verbraeck	TU Delft (TUD)	TPM, Multi Actor Systems	A.Verbraeck@tudelft.nl	tenured	–	Applicant

Dr Robert Verburg	TU Delft (TUD)	TPM, Values, Technology and Innovation	R.M.Verburg@tudelft.nl	tenured	–	Applicant
Dr Samantha Copeland	TU Delft (TUD)	TPM, Values, Technology and Innovation	S.M.Copeland@tudelft.nl	tenure-track	–	Applicant
Dr. Ir. Geertje Bekebrede	TU Delft (TUD)	TPM, Multi Actor Systems	g.bekebrede@tudelft.nl	Fixed position	–	Applicant
Professor Maartje Schermer	Erasmus MC (EMC)	Medical Ethics, Philosophy and History of Medicine	m.schermer@erasmusmc.nl	tenured		Applicant
Anouk van Alphen MSc	Erasmus MC (EMC)	Otorhinolaryngology and Head & Neck Surgery	a.vanalphen@erasmusmc.nl	Other (temporary contract)	Mix	Applicant
Dr Cobus van Wyk	Erasmus MC (EMC)	Integral Capacity management	j.vanwyk@erasmusmc.nl	other (temporary contract)		Applicant
Professor Suzanne van de Vathorst	Erasmus MC (EMC)	Medical Ethics, Philosophy and History of Medicine	s.vandevathorst@erasmusmc.nl	tenured		Applicant
Susanne Maassen MSc	Erasmus MC (EMC)	Quality and Patientcare	s.maassen@erasmusmc.nl			Applicant
Eline Krijkamp MSc	Erasmus MC (EMC)	Epidemiology	e.krijkamp@erasmusmc.nl			Applicant
Professor Erik van Raaij	Erasmus University Rotterdam (EUR)	RSM, department Technology & Operations Management; ESHPM, section Health Services Management and Organization	eraaij@rsm.nl	tenured	–	Applicant
Dr Sandra Sülz	Erasmus University Rotterdam (EUR)	ESHPM, section Health Services Management and Organization	sulz@eshpm.eur.nl	tenured	–	Applicant
Professor Kees Ahaus	Erasmus University Rotterdam (EUR)	ESHPM, section Health Services Management and Organization	ahaus@eshpm.eur.nl	tenured	–	Applicant
Dr Pieter Bakx	Erasmus University Rotterdam (EUR)	ESHPM, Health Economics group	bakx@eshpm.eur.nl	tenured	–	Applicant
Professor Pieter van Baal	Erasmus University Rotterdam (EUR)	ESHPM, Health Economics group	vanbaal@eshpm.eur.nl	tenured	–	Applicant
Dr Jan-Willem Weenink	Erasmus University Rotterdam (EUR)	ESHPM, Healthcare governance group	weenink@eshpm.eur.nl	tenured	–	Applicant

Corresponding Lead: Dr. Saba Hinrichs-Krapels
Planned start date of seed funding: 01-09-2022
Primary Convergence theme: Technology-supported transition in healthcare
Secondary Convergence theme: -
Tertiary Convergence theme: -
Quaternary Convergence theme: -
Keywords: resilience, prioritisation, capacity management, workforce, simulation modelling, serious games
please list a maximum of 7 keywords

1.b Summary (max 300 words)

The COVID-19 pandemic has highlighted the vulnerability of our healthcare systems, expressed in workforce scarcity, supply chain criticality, and patient allocation problems, combined with poor coordination and inadequate governance. Future abrupt or chronic stresses may cause similar or even more severe difficulties.

Our vision is to create a **Resilient Healthcare Hub** of research, education and valorisation activities that lead to resilient healthcare systems that can prepare, adapt, respond and learn from crises.

We aim to:

1. **Develop an empirically-derived patient prioritisation model** for use under scarcity conditions, and test its combined clinical, economic and ethical impacts.
2. **Design flexible and intelligent planning algorithms** for supplying care when resources are constrained.
3. **Develop governance processes** that ensure professional autonomy and safety culture under stress.
4. **Incorporate the social and organisational hospital context** within above models, algorithms and processes.
5. **Integrate planning modules** into hospital information system landscapes designed for robustness in emergency and normal situations.
6. **Develop and conduct awareness and training games** for hospital staff to prepare via different simulated stress scenarios.
7. **Expand and transfer scientific learning** to other hospitals and healthcare providers in the wider healthcare system.
8. **Prepare the next generation** of healthcare professionals, researchers and students in resilience preparedness.

We combine expertise in medical decision-making, surgery, health economics, mathematical programming, operations research, safety science, healthcare governance, medical ethics, organisation science, information systems research, complex systems engineering, simulation modelling, participatory research, and serious games. We apply and integrate methods from these disciplines to push scientific breakthroughs in **designing complex adaptive socio-technical systems, advancing health systems research, and resilience studies**. The **Resilient Healthcare Hub** will be an ecosystem for researchers, students and practitioners to transfer knowledge initially from resilience in a hospital setting to the broader healthcare system.

1.c Public Summary (max 300 words)

De COVID-19 pandemie liet zien hoe kwetsbaar ons zorgsysteem is. Er was een tekort aan personeel, beperkte hoeveelheid aan materialen en goede patiënten spreiding bleek een uitdaging. Ook was er sprake van onvoldoende coördinatie door suboptimaal bestuur en beleid. In de toekomst kunnen zich vaker acute of chronische crisissituaties voordoen die ons zorgsysteem in grote problemen kunnen brengen.

Wij willen door wetenschappelijk onderzoek en onderwijs toewerken naar een veerkrachtig zorgsysteem dat in staat is om zich voor te bereiden op crisissituaties, aanpassingsvermogen heeft en van deze crisissituaties kan leren.

Wij zullen:

1. Een model ontwikkelen dat patiënten kan prioriteren in tijden van schaarste. Vervolgens brengen we de financiële gevolgen in kaart, samen met het effect op klinische uitkomst en medisch-ethisch handelen.
2. Flexibele en intelligente nieuwe planningsmethodes ontwikkelen voor het leveren van zorg bij schaarse middelen.
3. Een nieuwe manier van ziekenhuizen besturen ontwikkelen die autonoom handelen van zorgprofessionals in crisissituaties vergemakkelijkt, waardoor zij zich betrokken en veilig voelen.
4. Bovenstaande samenbrengen in veerkrachtige, duurzame en praktische oplossingen voor normale situaties en crisissituaties, die rekening houden met de sociale en organisatorische context van ziekenhuizen.
5. 'Serious games' ontwikkelen en gebruiken. Dit zijn (computer)spellen waarin situaties worden nagebootst, om zorgprofessionals bewust te maken van en te trainen voor diverse crisissituaties.
6. De opgedane kennis en ervaring gebruiken om ook andere ziekenhuizen en zorgprofessionals verder te helpen en de volgende generatie zorgprofessionals en wetenschappers hierin op te leiden.

Ons consortium combineert expertise op het gebied van medische besliskunde, chirurgie, gezondheidseconomie, wiskundig programmeren, 'operations research', veiligheidswetenschappen, bestuur en beleid in de gezondheidszorg, medische ethiek, organisatiewetenschappen, ziekenhuisinformatiesystemen, 'systems engineering', simulatiemodellen, actie-onderzoek, en 'serious games'.

Samen streven we naar wetenschappelijke doorbraken en veranderingen in de praktijk op het gebied van een veerkrachtig zorgsysteem.

2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

Our starting hypothesis is that a comprehensive, resilient and value-based system is essential for sustainable, accessible and affordable healthcare. This was demonstrated by the COVID-19 pandemic during which governance, care coordination, and availability of critical supplies was below par, despite the innovation potential of many individual health and care professionals. Future stressors such as accumulating effects of financial constraints and burned-out workforce could result in worse outcomes.

If future healthcare is to provide **equitable access** to all in a sustainable way as envisioned by the Convergence mission, we need a different approach to organising care. While many initiatives focus on developing *technology solutions* to improve clinical care, this proposal addresses the organisational domain of care, for which **socio-technical solutions** are required. By doing so, we directly address the needs for a **“data-driven”, “value-based, resilient and inclusive care system”** (aims from the *Improving Health Journeys and Technology-based transitions in health(care) theme* of the Convergence agenda).

“Resilience is the ability of a socio-technical system (such as a healthcare system or a hospital) to sustain its key functions in response to shocks through absorbing, responding, recovering, adapting and learning¹. **Our mission is to enable healthcare organisations to be resilient in coping with future situations of scarcity via a Resilient Healthcare Hub**. Put simply, a hospital or other healthcare provider would have decision-making processes, protocols, and plans in place to cope with prioritising patients, managing its supplies, empower its workforce, and be adaptable and flexible in their response to crises.

To achieve this, our goals are to deliver *socio-technical* solutions:

- empirically-derived data-driven patient prioritisation model for use under scarcity conditions, tested for clinical, economic and ethical impacts;
- flexible and intelligent planning algorithms for supplying care, which we integrate into the planning modules of a hospital;
- governance processes that ensure professional autonomy and safety culture under stress;
- three awareness and training ‘serious games’ for hospital staff for preparing for different stress scenarios.

A key challenge for achieving resilience in the organisational domain of healthcare systems is the ability to take a cross-sectional and layered view of the different components that affect care delivery, and remove silos.² We therefore plan, via matched funding, to broaden our research to other hospitals and to the wider healthcare system, thereby responding to calls to create resilience across healthcare systems as a whole.³

To address the Convergence mission of **“tearing down boundaries between disciplines and institutions”**, we bring over 10 disciplines together, and co-lead our programme with healthcare practitioners and management staff. Moreover, our work has a within-project engagement, impact and valorisation plan, including commitments from practitioners, co-design and participatory approaches. This cross-disciplinary approach will help to finally resolve the long-standing planning issues in our complex healthcare system.

Finally, addressing the Convergence aim to “**interact, inspire, co-create, learn, and strengthen each other**”, we create a thriving and vibrant hybrid **Resilient Healthcare Hub** working space, where researchers, practitioners, decision-makers and students join in sharing ideas and playing serious games to learn about resilience both in theory and practice.

2.b Synergy between disciplines

In our consortium we merge disciplines into answering complex technical, social, economic and ethical questions that any health organisations face in trying to achieve resilience:

1. ***How can we manage care demand by prioritising patients during scarcity, and what ethical and cost implications does this have for our health services?*** We start our focus in one hospital as the selected healthcare organisation, Erasmus MC (EMC), as this is where pressures are felt during crises, and focus on **acute care** as our use case. Using expertise in **medical decision-making, surgical care, medical ethics** and **health economics**, we empirically demonstrate how relevant constraints (e.g., shortage of beds or personnel) affect health loss in acute Dutch hospital settings, and what the clinical, economic and ethical implications are of prioritising patients differently.
2. ***How can we be flexible in care planning and capacity management (i.e., planning workforce, beds and materials) in times of scarce constraints, when such planning tends to happen in departmental silos?*** Bringing expertise from **capacity management professionals, operations research, mathematical programming and meta-heuristics** combined with **qualitative research**, we model and develop flexible and adaptable strategies for supplying care when hospital resources are constrained, combining different data sets into hospital planning algorithms.
3. ***How can we enable our workforce, the backbone of our organisation, to display professional autonomy and a ‘safety culture’ during stress?*** Via expertise in **healthcare governance, safety science, healthcare quality, and leadership management** we use mixed **quantitative and action research methods** to enable the workforce to deal with crisis situations in responsive and responsible ways, combining the required innovation in these situations to deliver quality care.
4. ***Having developed resilient strategies for coping with stress above, how do we deal with the complex decision-making environment of the hospital?*** Using expertise in **organisational management, medical ethics, and complex systems engineering** we use **information flow modelling** and **participatory research** to identify the decision flows and operating procedures in the hospital, and **co-create strategies** to integrate learning from steps 1-3 into new ways of working in the hospital.
5. ***How can we integrate developed models above into intelligent planning algorithms within the hospital information system landscape?*** We bring **information systems research, algorithms** and **simulation modelling** and co-lead this work with **capacity management** practitioners within the hospital to integrate new planning and prioritisation models developed in 1-3 into the planning and scheduling of care within the hospital.

6. **How can any of our developed ‘resilient strategies’ be experienced by hospital staff and future researchers and students without waiting for another pandemic?** Using expertise in **complex systems engineering** and **serious game development**, knowledge and outputs from previous activities (on patient demand, care supply and culture) are integrated into viable simulation models for ‘how we operate under different scarce conditions’ from both a technical and social perspective, and used to develop awareness and training games.
7. **How can this learning on resilience continue to evolve and be transferred beyond EMC?** Learning from this initial flagship programme will form the first pilot project in our **Resilient Healthcare Hub**, where practitioners, students and researchers can learn about resilience, experience and co-develop simulation games, and spin-out into relevant projects that address resilience in other parts of the healthcare system (see Section on long-term sustainability).

The specific domain expertise and output will be integrated and subsequently feed into the training we will develop (Figure 1):

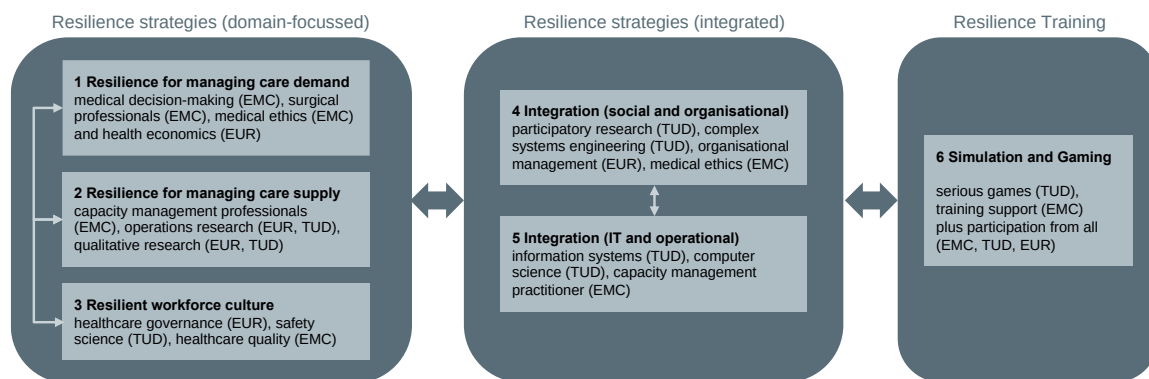


Figure 1. Overview of expertise and institutions involved

2.c Long term sustainability plan

The tenets of our feasibility and long-term sustainability plan are:

1. We start small and work incrementally: We deliberately take the use case of operating room (OR) patient prioritisation in the EMC as a comprehensive proof-of-concept to show how our research may contribute to the resilience of the wider healthcare system. We designed our work packages (Section 3c) to ensure a feasible demonstration of how resilience can be achieved in one organisation (EMC), ensuring cross-sectional aspects of the hospital (patient demand, care supplies and workforce) are integrated, and engaging practitioners into training and simulation games during the life of the project. This will result in domain-focussed resilience strategies and integrated resilience strategies for the whole hospital context (Figure 1).

2. We have enormous scale-up potential: Our goal is to expand beyond EMC to other hospitals. By awarded, submitted and future matched funding, we perform research on healthcare resilience in other hospitals and healthcare organisations and combine the results with the Flagship research. *This is bi-directional:* the matched funding projects will provide input, but also allow the scientific findings and developed outputs of the Flagship to be taken to the wider healthcare system. With this convergence application, we have already achieved our matched funding goal of **<1 million EURO** with secured awards relevant to this programme (Section 4a), and await results of grants submitted amounting to **<4 million EURO**.

Next potential topics of possible future funding and collaborations are:

- Explore feasibility and transferability of our findings to other hospitals and non-hospital settings
- Involve global partners to conduct international comparisons of similar initiatives in other health systems. We will connect our project to other, nationwide/international agendas (e.g., WRR, WHO and EU) which will remain top priority.
- Approach insurance companies who have an interest in agile patient prioritisation models.
- Explore the commercial valorisation potential of the serious games by sharing the modules with a commercial game company.

Potential target funds that fit our Flagship are:

- EU4Health 2021-2027 (One of their four targets is 'making healthcare systems stronger, more resilient and more efficient')
- Horizon Europe Disaster-Resilient Society (DRS)
- ZonMw 'Safety-II en veiligheidsergonomie' Other ZonMw and Horizon Europe programs on quality and safety in healthcare
- Dutch Research Council (NWO) Dutch Research Agenda Research along Routes by Consortia (NWA-ORC)
- NWO Open Technology Program
- ERC starting grant, NWO Vidi

3. Buzz-words will change, but relevance continues: Resilience is a concept with long-term relevance.^{4,5} Although the focus on COVID-19 will probably diminish in the coming years, it is just one example of an incidental stressor on healthcare systems globally. It revealed underlying

vulnerabilities which were already present and are likely to become more so in the future. Creating algorithms, models and strategies for being 'resilient' can only create benefits for furthering and creating more efficient and sustainable healthcare in the future. These incremental steps to ensure 'resilience' of healthcare systems through our flagship will therefore have long lasting effects on the system as a whole, which is why it has been regarded as a priority by the WHO in achieving Sustainable Development Goals (SDGs).⁶

Specific **project risks** are stated in 3b.

2.d Visibility, attracting talents and new partners

The purpose of creating a **Resilient Healthcare Hub** is to generate enthusiasm for the flagship, bring together individuals with a unified goal of resilience in healthcare systems, and create visibility for future partners (researchers, practitioners and students). At project start, the Hub would consist of 34 initial members (26 consortium co-applicants, 8 new hires) plus 8+ MSc students researching related topics for their graduation theses, which we expand as follows:

Clinical partners: We have ongoing collaborations with several hospitals in the Netherlands (Maasstad Hospital, LUMC, HagaZiekenhuis, OLVG, Amsterdam UMC, ETZ, Amphia, and Sfg). Practitioners from these hospitals will be invited to participate in the serious games and train their personnel within the **Hub**, further bolstering our nationwide visibility among clinicians. We will continue publishing regular updates on our scientific progress on our website and in professional magazines which are well-read by clinicians (e.g. Medisch Contact).

Wider partners: Hospital partners above, plus new collaborations with non-hospital providers will allow us to pursue more ambitious funding opportunities. Building on our existing awarded matched funding (Section 4a), we will jointly apply for grants, targeting larger ambitious funding opportunities that require multiple partners (e.g., NWA-ORC, Horizon Europe).

Academic and non-academic dissemination: Scientific outputs of our project will be disseminated via peer-reviewed articles, (inter)national conferences and policy reports, exclusively in open access. To reach a wider audience, social media and networks (e.g. LinkedIn, ResearchGate, Twitter) will be used to share our progress and preliminary results. A **Resilient Healthcare Hub** website will help us reach a wider community of interest. Our ideas, progress and involved partners will be posted and with our blog, we will engage in discourse with other researchers. We are already involved in various research networks (e.g. Resilience in Healthcare Research Network, for a detailed overview see 4.c), which allows us to gain valuable feedback.

Teaching and attracting new talents: Further to targeted education activities (Section 4a), we will:

- Promote 'combined teaching' (e.g. lectures given by academics from different institutions). This symbolises our interdisciplinary collaborations and shows students the added value of Convergence.
- Supervise shared PhD candidates, who will subsequently supervise shared BSc/MSc students. In this way, the Convergence will go through the 'generations'. New, young academic talents

will be involved into interdisciplinary activities from the beginning with preliminary modelling work

- Provide a unique opportunity for TUD/EUR MSc students to conduct thesis projects within a real hospital setting, with access to data and direct connections to healthcare professionals, putting their theory into practice.
- State-of-the-art skills training for medical students with our serious games, complemented with teaching on value-based healthcare, medical decision-making and organisational aspects of the healthcare system, which up till now is a blind spot in the curriculum
- Create a **Student Hub Club** embedded within the Hub, dedicated to young researchers with freely accessible articles on resilience, student/PhD blogs, interesting conferences to attend, and vacancies for thesis projects.

2.e Long term societal impact

Our 'North star' is to improve the resilience of healthcare systems, characterised by the ability of any healthcare provider to anticipate, bounce back from unknown unknowns, demonstrating flexibility and agility in its response to crises, including the regular stresses in healthcare. It also means fully integrated planning within a single hospital and across other healthcare providers in the system, to coordinate activities and efficiently allocate resources during different demand surges.

We acknowledge the incremental steps required towards this ambitious goal (Figure 2), where we demonstrate how our flagship programme contributes to direct outcomes within its 5 years, and long-term impact.

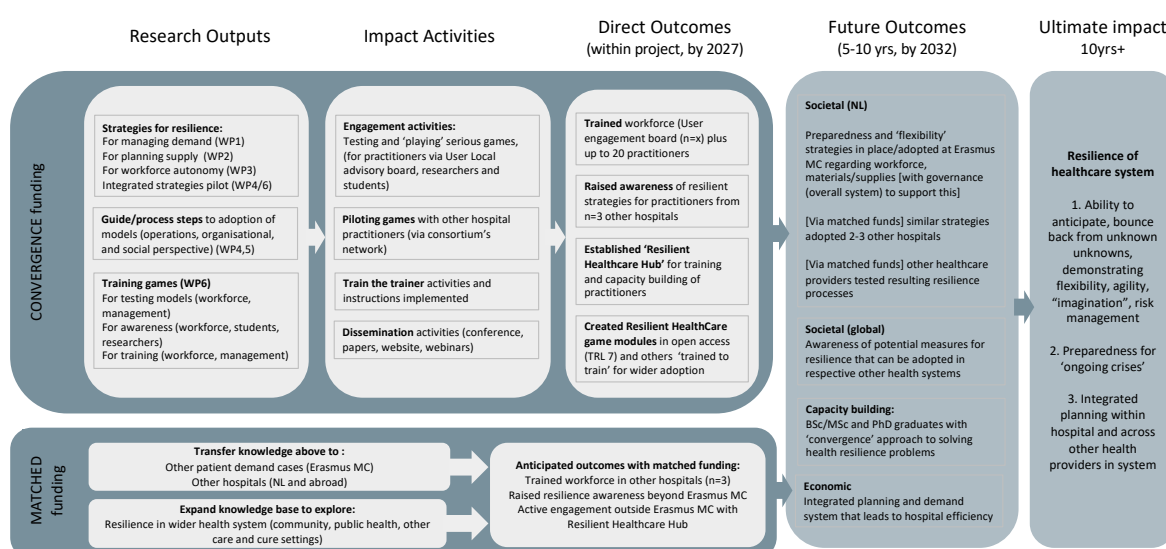


Figure 2: Societal and economic impact (WP stands for Work Packages described in Section 3c)

The initial research outputs (patient prioritisation model, intelligent planning algorithms, governance processes) will ignite the development of the serious games and workshops, which are aimed to test, demonstrate and train the added value of integrated planning methods. These games are both research outputs but also serve as engagement activities:

1. **Awareness games and workshops** show the potential of using integrated planning, and its effects on social, economical, medical and ethical values (aimed at healthcare professionals and students; used in MSc programs of EUR, EMC and TUD).
2. **Simulation games** help to assess the effects and potential issues of integrating demand planning, materials planning, and staff planning. It will use regular and crisis scenarios to test and demonstrate the effects in co-creation workshops with health professionals (WP4), and later tested for integration within the hospital planning platform (WP5).
3. **Training** for hospital planners ready for deployment in the hospital.

All three gamified workshops will have Technology Readiness Level (TRL) 8-9, and will be ready for full deployment during and after the project. Members of the User Board (Section 4c for committed list of individuals), MSc/BSc students and consortium researchers will be involved in developing, piloting and playing the games. We have engaged this User Board in writing this proposal; and ensured we are aligned with the annual strategy plan of EMC and the Medical Training curriculum for 'future proof clinicians'. We envisage 30 EMC practitioners to have been trained through the serious games during the project. We will invite practitioners from other hospitals to take part in piloting the games, raising their awareness of the potential to adopt resilience strategies. We intend to commercialise the games and integrated planning systems.

All the above will be delivered during the 5 years of the project via Convergence funding. Matched funding activities include transfer of knowledge to other hospitals nationally and globally, and exploring how to apply the learned concepts of resilience within the wider healthcare system (Figure 2).

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

Day-to-day activities in healthcare systems often fall short of what is technologically possible, caused by crippling funding cuts, workforce burnouts, inefficient drug and equipment supply chains, and poor forecasting for capacity needs. These issues became apparent in the COVID-19 crisis, but are also present in everyday challenges. Addressing socio-technical challenges within the organisational domain of healthcare systems, as is the focus of our flagship, is of crucial importance to enable healthcare systems globally to function in any conditions creating scarcity of resources, such as pandemics and disasters, as well as day-to-day stresses on the system.

Following COVID-19, the World Health Organisation (WHO) made policy recommendations for increasing health systems resilience, and pointed to the importance of addressing resilience in all aspects of the healthcare system,^{3,6} including transforming 'obsolete' practices or redesigning services, with implications for the workforce.

However, empirical research and resulting guidance that enables a health organisation to learn *how* to become resilient and prepared for conditions of scarcity are lacking.¹ Innovations that take into account the complexity of the healthcare system are needed, and while these can be designed in theory or in simulation models, embedding them into organisations requires further innovations.

Our ambition within this **Resilient Healthcare Hub** is to **enable healthcare organisations to become resilient in coping with future situations of scarcity**. We address gaps in scientific research and practice:

a. We respond to global calls for evidence in resilience studies: Contrary to previous studies⁷, we will integrate systems approaches that are adaptive and responsive to care needs, can deal with uncertainties in surge and demand⁸, and consider relationships between the behaviour of individual health providers and wider policy decisions.⁹

b. We directly tackle 'tough choices' needed for resilience: Dealing with scarce financial, human, and material resources means making tough choices, including the ethical dilemmas of prioritising patient (groups).¹⁰ These choices are our starting point by building an empirical evidence base on the consequences of choices for clinical, ethical and cost outcomes.

c. We overcome existing silos in healthcare practice: We study combined effects of patient prioritisation (managing demand) and capacity management (managing supplies), especially taking into account workforce planning under different stress scenarios, responding to existing silos and fragmentation in how organisations are managed.¹¹

d. We bridge disparate scientific disciplines: We bridge expertise in medical decision-making, health economics, mathematical programming, operations research, safety science, healthcare governance, medical ethics, organisation science, information systems research, complex systems engineering, simulation modelling, participatory research, and serious games. We apply and integrate methods from these disciplines to push scientific breakthroughs in designing complex adaptive socio-technical systems, advancing health systems research, and resilience studies.

e. We advance empirical research on resilience that is directly applicable in daily practice:

We pave the way for real-life applicability of our models by creating a simulation lab (flex space of the Otorhinolaryngology department), where hospital professionals can engage and 'train' for different modes of working, dealing with different scenarios, and where the full consortium and community can meet for co-creation workshops and presentations in addition to the online **Resilient Healthcare Hub** presence. Through action research, we will test our hypotheses and strategies with our own workforce in a real-world environment.

f. We plan to expand beyond the acute hospital setting: We design scenarios that are applicable not just in a major disruption but also in conditions of general scarcity such as financial constraints. This makes the simulation training attractive for general resilience capacity building for many hospitals and other healthcare providers nationally and globally. Future projects can explore interdependencies and links between health providers in times of crises.

g. We pave the way for future innovation in care organisation (AI, complexity, modelling): Most existing AI-health research focuses on personalised care, rather than on the organisational domain of care. Techniques and empirical work developed in this proposal (e.g, data-driven mathematical programming, meta-heuristics, and agent-based simulation) creates opportunities to bring machine learning algorithms to the much-neglected organisational domain of care¹².

3.b Aims and feasibility

To achieve our mission of enabling healthcare organisations to be resilient in coping with future situations of scarcity, we will:

1. **Develop an empirically-derived patient prioritisation model** for use under scarcity conditions, and test its combined clinical, economic and ethical impacts.
2. **Design flexible and intelligent planning algorithms** for supplying care when resources are constrained.
3. **Develop governance processes** that ensure professional autonomy and safety culture under stress.
4. **Incorporate the social and organisational hospital context** within above models, algorithms and processes.
5. **Integrate planning modules** into hospital information system landscapes designed for robustness in emergency and normal situations.
6. **Develop and conduct awareness and training games** for hospital staff for preparing for different simulated stress scenarios.
7. **Expand and transfer scientific learning** to other hospitals and healthcare providers in the wider healthcare system.
8. **Prepare the next generation** of healthcare professionals, researchers and students in resilience preparedness.

To ensure **feasibility**, activities are designed to run immediately, building on prior research (Section 4a), a funded Convergence Open Mind project and an Erasmus MC Koers23 grant. We secured **>€1M** in awarded grants for the first 2 years and **>€4M** of possible matched funding from submitted grants

(Section 4a). Students will be involved in MSc projects (Section 3c), engaged in the Student Hub Club (Section 2d) and development of serious games and training materials **TU Delft game lab**.

Table 1. Project risks with proposed mitigation strategies (only 'Medium/High' likelihood/impact included):

<i>Risk</i>	<i>Mitigation</i>
Insufficient data for novel planning algorithms	Direct involvement of EMC, CBS and DHD provides data access or expert estimates
Resistance of hospital staff to change	WP3-4 assesses barriers and involves professionals
Good planning models but invalid for organisational implementation	WP3-4 has co-creation workshops with practitioners
Integration impossible between supply/demand/staffing models	WP6 simulates integration early for adaptations in WP1-3
No uptake of project deliverables	3 hospitals involved in games for knowledge transfer; WP4 workshops facilitates uptake
Technical barriers for integration with Hospital Information Systems	Early involvement of IT staff from hospitals

3.c Work plan

Coherence of the programme is ensured by aligning activities under the **Resilient Healthcare Hub** banner. Its hybrid virtual-physical mode enables global partners to engage and makes use of the **infrastructure** facilities at EMC (e.g. office space, Research Suite) and TU Delft game lab¹³.

Required data sources include EMC administrative and capacity management data, hospital materials purchase history/database, Netherlands Statistics (CBS), Dutch Hospital Data (DHD), ValueCare, national database surveys (e.g., worker satisfaction, absenteeism), hospital protocols.

The activities in the workpackages (WP) 1-3 will be integrated in WPs 4-6, and via matched funding extended to other hospitals and the wider healthcare system (Figure 3).

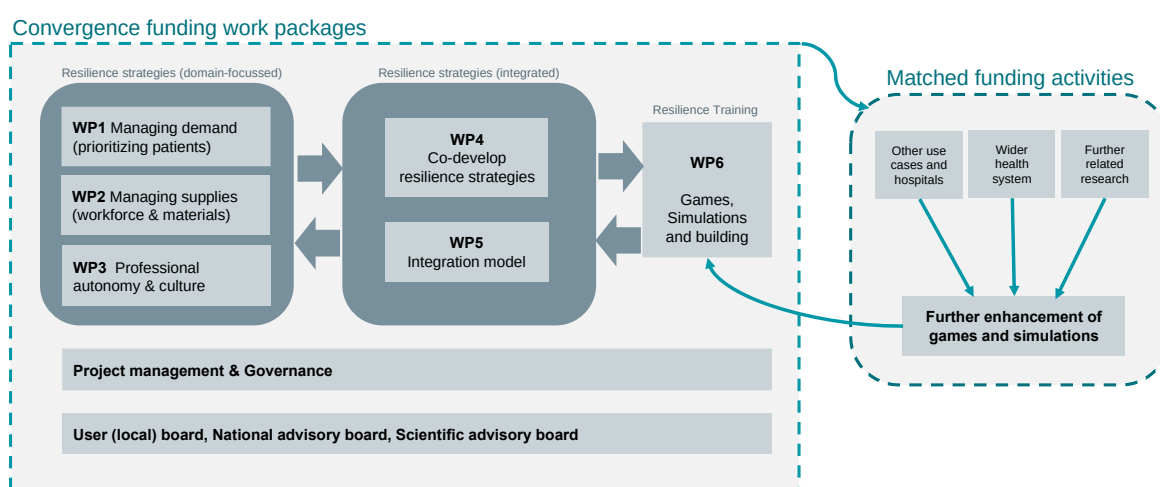


Figure 3: Overview of work packages and long-term activities via matched funding

WP 1: Managing demand (OR) patient prioritisation)

Resources: MSc students, 1 shared PhD student (EMC/EUR), years 1-4

Matched: €22.334 awarded grants, €868.548 submitted grants

Aim: Develop an empirically-derived patient prioritisation model for the OR for use under scarcity conditions, and test its combined clinical, economic and ethical impacts.

Tasks:

- Extend previously-developed decision model that quantifies consequences of surgical delay in terms of expected health loss¹⁴ to alternative treatment options (thereby transforming from 'surgical intervention based' to 'diagnosis based')
- Determine optimal treatment and diagnostic strategies in various scenarios (interfaces WP6); the absence or presence of various constraints (interfaces WP2).

Methods: Statistical and econometric analysis (e.g. Markov model) of internal clinical and registry data, literature data, expert opinions elicitation.

WP2: Managing supplies (workforce and materials)**Resources:** MSc students, 1 shared PhD candidate (TUD/EUR), years 1-4**Matched:** €198.917 awarded grants, €1.231.299 submitted grants

Aim: Design 'flexible' strategies for supplying care when resources (workforce, materials, beds) are constrained.

Tasks:

- Map relevant stakeholders, data sources and protocols for planning workforce shifts, logistics and materials management at EMC hospital (interfaces WP3,4).
- Model 'flexible and adaptable' strategies for planning work shifts, patient flows (logistics), and materials (equipment and commodities), and investigate their interdependencies and constraints together with demand models (from WP1).¹⁵
- Simulate different scenarios (shortages in one/multiple type(s) of resource) and strategies to predict the expected effects on availability and quality of care (interfaces WP6)

Methods: qualitative interviews, document analysis; mathematical programming model solved with meta-heuristics; evaluated by discrete-event simulations.

WP3: Professional autonomy and safety culture**Resources:** MSc students, 1 shared PhD candidate (TUD/EUR), years 1-4.**Matched:** €478.000 awarded grants, €20.000 submitted grants (see section 4a)

Aim: Develop governance processes that ensure professional autonomy and safety culture under stress.

Tasks:

- Identify international good practices
- Conduct ethnographic research on existing culture and practices, using Safety II framework¹⁶ to guide analysis
- Test interventions at team and department level to increase collective professional autonomy while remaining accountable, using concepts of 'reflexive spaces'¹⁷.

Methods: review of international practices; mixed methods: ethnographic observations, (group) interviews, and questionnaires.

WP4: Integrating new models into the hospital (social and organisational perspective)**Resources:** MSc students, 1 shared PhD candidate (TUD/EUR/EMC), years 1-4.**Matched:** €184.000 awarded grants, €1.479.000 submitted grants (see section 4a)

Aim: Incorporate the social and organisational hospital context within models, algorithms and processes (from WPs1-3).

Tasks:

- Map stakeholders involved in decision-making processes required for models and strategies from WP1-3

- Capture interdependencies of decisions and hospital information flows and how they are impacted by the models and decision support tools in WPs 1-3.
- Identify critical information for decision-making during different scarce scenario situations, along with the required information and information sources (e.g., hospital protocols and standard operating procedures)
- Perform ethical analysis of both the implicit values underlying the model and the (expressed) views on professional values of hospital staff.

Methods: document content analyses, conceptual modelling and information flow mapping, qualitative interviews, participatory workshops

WP5: Integration into hospital planning system and workflow

Resources: Postdoc (EMC); years 3-5; IT developer (EMC)

Matched: €120.000 awarded grants, €799.000 submitted grants (see section 4a)

Aim: Integrate planning modules into hospital information system landscape designed for robustness in emergency and normal situations.

Tasks:

- Study current planning modules in the Hospital Information System Landscape (HISL) of the project partners; assess system vulnerabilities during critical events.
- Functional design to integrate the planning methods developed in WP1-3 into the existing HISL and workflows.
- Prototype implementation of the algorithms from WP1-3 and interfacing them to the wider HISL.
- Prototype testing and demonstration in a mock HISL environment, in cooperation with HISL vendors, to study its effectiveness in simulated regular and crisis situations.

Methods: T5.1-4: Scrum framework¹⁸, empirical optimisation, testing frameworks.

WP6 Integrated simulation and gaming for testing and training

Resources: Postdoc (TUD); years 3-5; game development budget

Aim: Develop and conduct awareness and training games for hospital staff for preparing for different simulated stress scenarios.

Tasks:

- Develop an integrated simulation model to assess and demonstrate the combination of demand, supply, and staff planning for resilience scenarios.
- Develop and apply a 2-hour awareness game (tabletop/digital) on integrated planning methods for healthcare professionals and students.
- Develop and apply a half-day exploration game for WP4 to test resilience scenarios by healthcare professionals and management.
- Develop a 2-day training game for adopting the resilient planning strategies and software by healthcare planners.

Methods: mixed discrete-event, agent-based simulation. Triadic Game Design method; observations; surveys.

3.d Alignment of the program with relevant national and international research agendas

Recently, the Netherlands Scientific Council for Government Policy and the Royal Netherlands Academy for Arts and Sciences published advice on 'navigating and anticipating in uncertain times',¹⁹ stressing need for broad societal resilience and anticipate a variety of scenarios. We also align with the 'routes' of the Dutch Research Agenda: 'towards resilient societies'²⁰ Lastly, the Government of the Netherlands has also established a program 'New chances for Topsector Life Sciences & Health' which incorporates a 'pandemic preparedness' initiative.

Internationally, resilience in healthcare has gained much attention internationally leading to new (research) programs (e.g., Horizon Europe funding calls on resilience, journal special issues¹. EU4Health 2021-2027 has the ambition to "strengthen health systems, their resilience and resource efficiency"²¹ This also goes for the European University Hospital Alliance. Moreover, our proposal targets crucial gaps in literature: a lack of empirical research on resilience,²² lack integrated of hospital planning and capacity management¹¹, and lack of research on governance.²³ Further, we directly address the for mixed methods approaches and investigating the ways in which different layers of the healthcare system interact to understand, and intervene within current practices of organising care.²⁴

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

Consortium members have a **strong track record in publishing** on specific topics of the programme, including health systems strengthening^{25,26}, surgical planning & capacity modelling^{27,28} [+ref], organisational culture^{29,30,31}, complex modelling³² [+ref], operations and supply management^{33,34}, materials management in hospitals^{35,36}, cost-effectiveness in hospitals^{37,38}, decision making in healthcare^{39,40}, implementing decisions support systems in healthcare⁴¹, and serious games⁴².

Membership of boards & working groups include chairing SAPEA Working Group for the ERR on the EU's future strategic crisis management, Vice-chair of the Clinical Governance working group at the Health Standards Organization in Canada, ZonMw board on the topic of Safety in Healthcare, Association for the Advancement of Artificial Intelligence (AAAI) and editorial board memberships of various scientific journals, including invited guest editor of Sustainability: Special Issue on Resilience and Sustainability of Health Systems.

Consortium members are able to **translated scientific results into clinical applications or societal impact**. Examples include:

General public

- Media exposure in (inter)national media, including Elsevier, Trouw, NPO, RTL Nieuws and ABC news
- Debate game to introduce the general public into different perspectives on OR scheduling in times of scarcity. Awarded as a Science Gallery project.
- TEDx Talk: How to talk about health care costs⁴³

Clinical impact

- Development of an online accessible tool for users (healthcare professionals) to interactively play with the OR triage decision model and compare surgeries⁴⁴
- Contributor to the Dutch Journal of Medicine (NTVG) panel on implementation of health care services.⁴⁵

Scientific audience

- Contributor to OECD/WHO report (2021) on elderly care
- Witness to the Australian Royal Commission on Aged Care Quality and Safety (2020)
- Participation in the nation-wide CEA Watch trial, resulting in a change in the guidelines
- Developed free online tools enabling researchers to estimate indirect medical cost. [ref]

Policy making

- Developed new supervision methods with the healthcare inspectorate that allow healthcare organisations to become more resilient. The results have changed regulatory practice.
- Led the design and delivery of policy labs that brought together policymakers, practitioners, and researchers to engage on how to bring research closer to scientific impact, in health policy domains⁴⁶.
- Changed guidelines for economic evaluations in healthcare.⁴⁷

BSc/BA and MSc/MA thesis supervision

Together, we supervise ~100 BSc/BA and MSc/MA students each year from EMC, EUR, TUD, the NFU MSc Quality & Safety and other universities, e.g. Health Sciences students. Many students have done or are currently on a project on resilience healthcare topics such as “Creating a flexible workforce”, “Long-term multi-priority surgery scheduling”, “OR scheduling considering downstream resources”, and “Resilient procurement strategies during pandemics”.

BSc/BA and MSc/MA teaching related to the project

We transfer our knowledge to and interact with thousands of students by teaching in more than 20 courses for BSc and MSc students in medicine from EMC, various BSc and MSc students from the EUR and TUD, the NFU master Quality & Safety and students from other universities. Topics of these courses include health systems research, resilience engineering, safety and security management in healthcare, quality and safety of healthcare, value-based healthcare, prognostic modelling, healthcare procurement & value chain management, health service operations management, operations research and serious gaming.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The leads of this project form a perfect fit with the Convergence goals, project aims and with each other:

- The main lead, Dr. Hinrichs-Krapels (TUD), is an expert on health systems engineering and policy. She has expertise in delivery of care in intramural settings, e.g. organisational strategies for improving efficiencies in hospitals and primary care.
- The lead from EUR, Prof. Bal, is an expert on healthcare resilience and healthcare governance. He works with the healthcare inspectorate and several hospitals to increase resilience and develop a just culture in hospitals and other healthcare organisations and is part of an international consortium to study resilience in healthcare.
- The leads from EMC, Prof. Baatenburg de Jong succeeded by Prof. Lingsma, have already been combining their clinical experience and experience in clinical decision making in a successful collaboration on decision modelling and OR capacity management. Being a specific use case, their research will form the starting point of the project.

Table 2 shows WP expertise delivered by consortium members:

Table 2. Expertise per WP

Expertise needed	Co-applicants delivering this expertise
Aim WP1: Develop an empirically-derived patient prioritisation model for use under scarcity conditions, and test its combined clinical, economic and ethical impacts. Co-leads: Lingsma (EMC), van Baal (EUR)	
Medical decision making / modelling	Lingsma, Stankova, Krijkamp
Surgery	Baatenburg de Jong, van Alphen
Cost-effectiveness/health economics	Baatenburg de Jong, Verhofstad, Van Baal, Bakx
Medical ethics	Schermer, Van de Vathorst
Aim WP2: Design 'flexible' strategies for supplying care when resources (workforce, materials, beds) are constrained. Co-leads: Van Essen (TUD), Sülz (EUR)	
Mathematical programming	van Essen
Meta-heuristics	van Essen
Optimising healthcare processes	Van Essen, Yorke-Smith
Purchasing and supply management in healthcare	van Raaij
Healthcare operations management	Sülz
Integral Capacity management	van Wyk
Materials management in healthcare	Hinrichs-Krapels
Process mapping; qualitative methods	Hinrichs-Krapels, van Raaij
Aim WP3: Develop governance processes that ensure professional autonomy and safety culture under stress. Co-leads: Bal (EUR), Grossman (TUD)	
Healthcare governance	Bal
Organisational (safety) culture	Bal, Grossmann, Weenink, Groeneweg
Quality of healthcare	Weenink, Hazes,
Leadership and management	Verburg
Action research; ethnography	Bal
Aim WP4: Incorporate the social and organisational hospital context within above models, algorithms and processes. (Co-leads: <i>Hinrichs-Krapels (TUD), Ahaus (EUR), Van de Vathorst (EMC)</i>)	
Complex systems engineering	Hinrichs-Krapels, Comes
Process and information flow mapping	Hinrichs-Krapels, Comes
Healthcare operations management	Ahaus
Medical ethics	Van de Vathorst, Schermer, Copeland, Van Busschbach, Van Alphen
Technology ethics	Copeland
Aim WP5: Integrate planning modules into hospital information system landscape designed for robustness in emergency and normal situations. (Co-leads: <i>Yorke-Smith (TUD), van Wyk (EMC)</i>)	
Operations research	Yorke-Smith, Van Essen
Capacity management and medical informatics	Van Wyk
Planning, resource allocation and supply chain management	Verbraeck

Integrated simulation models for complex systems	Verbraeck
Optimising healthcare processes	Van Essen
Algorithms	Yorke-Smith
Complex modelling	Yorke-Smith
<i>Aim WP6: Develop and conduct awareness and training games for hospital staff in different simulated stress scenarios. (Lead Bekebrede (TUD), Baatenburg de Jong (EMC))</i>	
Resilience (in healthcare)	Copeland, Hinrichs-Krapels
Developing serious games	Bekebrede, Verbraeck, Dankbaar
Embedding games into training for health professionals	Dankbaar, Baatenburg de Jong

4.c Involvement of partners

We have incorporated our stakeholders from the beginning into our project by means of three levels of advisory boards:

User board

The user board consists of stakeholders within the EMC. Already in the board are the heads of the department of Internal Medicine Integral Capacity Management (A Breedijk), Head of Intensive Care (D Gommers), Head of Surgery department (J Hendriks), Purchasing Department (M Timmerman), a developer of the OR decision model and a member steering committee EUHA (R Steensma). More people will be involved later. These people will be closely involved, meeting and/or giving advice 6 times per year. The goal of this board is to discuss current progress of the project, provide the consortium members with input (e.g. clinical, strategic or data), garner valuable feedback on practical assumptions made in different WPs and thereby identify gaps and areas with need for action, helping to fit with the needs and clinical practice of the users, align our project with long-term organisational goals for EMC, and create support and shared ownership to facilitate the use of our results in practice.

National advisory board

This board consists of stakeholders of (healthcare) organisations within the Netherlands, e.g. other hospitals, professional associations and regulatory bodies. The board already includes stakeholders from ROAZ South West Netherlands, Dutch Healthcare Institute, Dutch Association of Medical Specialist (FMS), Healthcare Inspectorate and Maasstad Hospital and will be complemented with people from other organisations. This board will meet and/or give advice at least twice per year. The goal of the board is to provide the consortium with strategic guidance on how to keep this project aligned with the ongoing national situation and transformations in healthcare. For the expansion of the current use case to other hospitals and healthcare systems we also use this national advisory board.

International scientific advisory board

This board consists of international scientific experts. This board will meet and/or give advice at least once a year. The goal of this board is to fortify the scientific base of the project by providing the consortium with an objective, external perspective on research goals and methods. The board

will review the quality and relevance of the project and address broad scientific matters on the topic of resilience studies. We will collaborate with members of this board to attract international funding.

These three boards are flexible structures and meant to be growing along with the growth of the Flagship consortium and project; during the project we will identify and involve more partners who we will invite to our boards.

In addition to the advisory boards, we are very well connected to a wide variety of external partners in the ecosystem (Figure 4) We will continue the collaborations and connections to these parties and grab opportunities when suitable for input and advice, specific collaborations, together applying for grants, matched funding by these organisations.

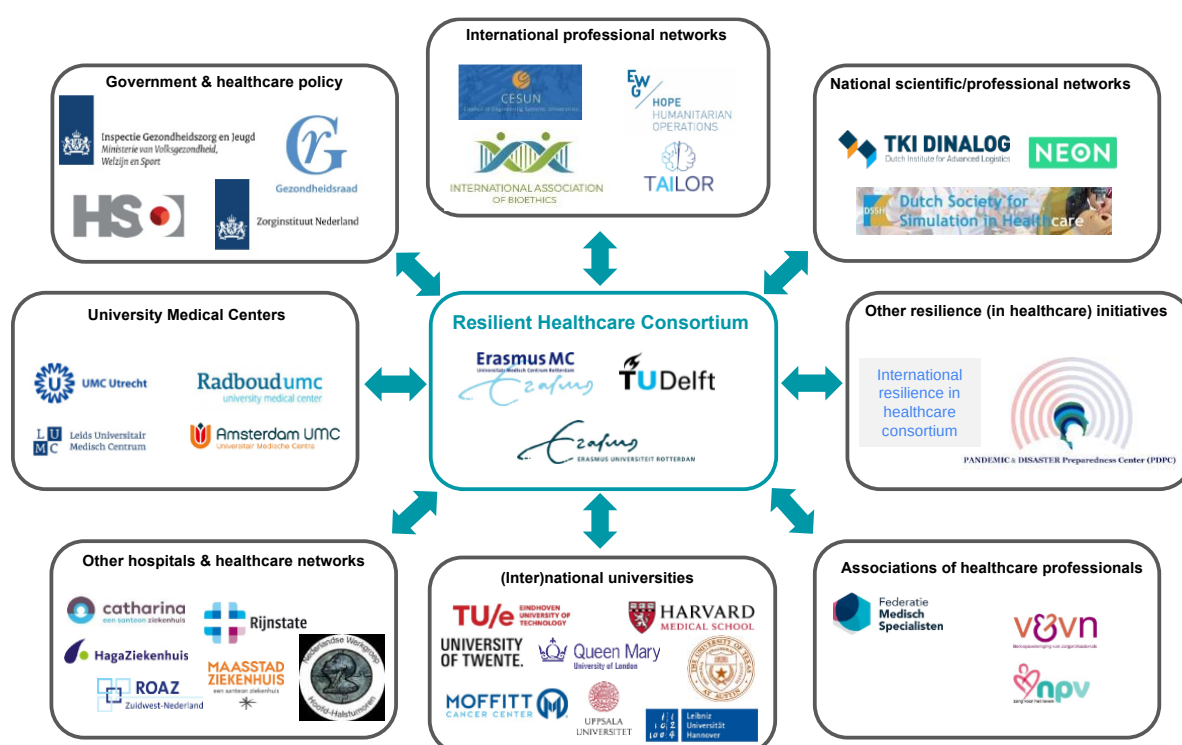


Figure 4 External connections

4.d Governance, organization and collaboration strategy of the consortium

Convergence by sharing

All WPs have shared goals and leads (Figure 5). Hired PhD students and postdocs have an official base, but are shared by the three institutes in practice, which we effectuate by having multiple supervisors for each PhD student and postdoc from the different institutes. 'Student Hub Club' will meet once a month.

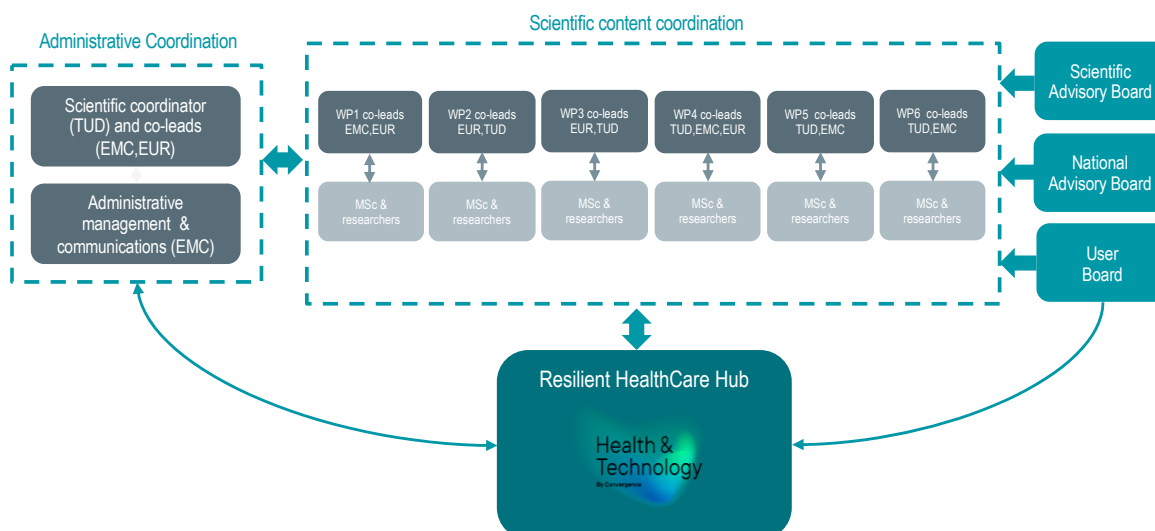


Figure 5 Organogram

Governance

We will have a management team, consisting of the project leads and 0,2 FTE managing assistant. This team will meet every two weeks, and will:

- Monitor, evaluate and re-direct the direction of the project and budget expenditure
- Monitor and facilitate the progress of the project, well-being of the postdocs and PhD candidates
- Monitor and stimulate the involvement of participants, advisory board members and external partners
- Initiate and organise writing of new shared external grant proposals
- Stimulate to identify talents among BSc/BA/MSc/MA students
- Facilitate talented hired PhD students and postdocs in their future careers (e.g., tenure tracks).

Meetings

Participants of the consortium will meet regularly to stimulate close connections and teamwork, hired PhD students, postdocs and IT developers will work for at least one day jointly at the EMC at the flex space of the Otorhinolaryngology department. Complete consortium will meet 3 times a year. We will hold bi-monthly interactive workshops, seminars and Q&A sessions, inviting senior researchers, and PhD students and postdocs to learn each other's language, terminology and research paradigms.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

IV References

¹ Wiig, S., & O'Hara, J. K. (2021). Resilient and responsive healthcare services and systems: challenges and opportunities in a changing world. *BMC Health Services Research*, 21(1), 1-5.

² van de Bovenkamp, H. M., Stoopendaal, A., & Bal, R. (2017). Working with layers: the governance and regulation of healthcare quality in an institutionally layered system. *Public Policy and Administration*, 32(1), 45-65.

³ World Health Organization (2021). Building health systems resilience for universal health coverage and health security during the COVID-19 pandemic and beyond: WHO position paper. Available at: <https://www.who.int/publications/i/item/WHO-UHL-PHC-SP-2021.01>

⁴ Haldane, V., Ong, S. E., Chuah, F. L. H., & Legido-Quigley, H. (2017). Health systems resilience: meaningful construct or catchphrase?. *The Lancet*, 389(10078), 1513.

⁵ Kieny, M. P., & Dovlo, D. (2015). Beyond Ebola: a new agenda for resilient health systems. *The Lancet*, 385(9963), 91-92.

⁶ Ziglio, E. (2017). Strengthening resilience: A priority shared by health 2020 and the sustainable development goals. Copenhagen: WHO Regional Office for Europe, 17.

⁷ Komashie, A., Ward, J., Bashford, T., Dickerson, T., Kaya, G. K., Liu, Y., ... & Clarkson, P. J. (2021). Systems approach to health service design, delivery and improvement: a systematic review and meta-analysis. *BMJ open*, 11(1), e037667.

⁸ Roemeling, O., Ahaus, K., van Zanten, F., Land, M., & Wennekes, P. (2019). How improving access times had unforeseen consequences: a case study in a Dutch hospital. *BMJ Open*. 9 [e031244]. doi: 10.1136/bmjopen-2019-031244.

- ⁹ Wiig, Siri, Karina Aase, and Roland Bal. 2021. "Reflexive spaces: Leveraging resilience into healthcare regulation and management." *Journal of Patient Safety*. doi: 10.1097/PTS.0000000000000658.
- ¹⁰ WRR report. Available at: <https://www.wrr.nl/publicaties/rapporten/2021/09/15/kiezen-voor-houdbare-zorg>
- ¹¹ van der Ham, A., Boersma, H., van Raak, A., Ruwaard, D., & van Merode, F. (2019). Identifying logistical parameters in hospitals: does literature reflect integration in hospitals? A scoping study. *Health services management research*, 32(3), 158- 165.
- ¹² Garg, A., Venkataramani, V. V., Karthikeyan, A., & Priyakumar, U. (2022, January). Modern AI/ML Methods for Healthcare: Opportunities and Challenges. In *International Conference on Distributed Computing and Internet Technology* (pp. 3-25). Springer, Cham.
- ¹³ TU Delft serious gaming facilities. Available at: <https://seriousgaming.tudelft.nl>
- ¹⁴ Gravesteyn B, Krijkamp E, Busschbach J, Geleijnse G, Helmrich IR, Bruinsma S, et al. Minimizing Population Health Loss in Times of Scarce Surgical Capacity During the Coronavirus Disease 2019 Crisis and Beyond: A Modeling Study. *Value Health*. 2021;24(5):648-57.
- ¹⁵ A.J. Schneider, M. Carlier, J.T. van Essen, E.W. Hans. Scheduling surgery groups considering multiple downstream resources. *European Journal of Operational Research* 282 (2): 741-752, 2020.
- ¹⁶ Hollnagel, Erik. 2013. "Making health care resilient: from safety-I to safety-II." In *Resilient Health Care*, edited by Eric Hollnagel, Jeffrey Braithwaite and Robert L. Wears, 3-18. Farnham: Ashgate.
- ¹⁷ Wiig, Siri, Karina Aase, and Roland Bal. 2021. "Reflexive spaces: Leveraging resilience into healthcare regulation and management." *Journal of Patient Safety*. doi: 10.1097/PTS.0000000000000658.
- ¹⁸ Michal Hron, Nikolaus Obwegeser: Why and how is Scrum being adapted in practice: A systematic review. *J. Syst. Softw.* 183: 111110 (2022)
- ¹⁹ WRR publicaitons. Available at: <https://english.wrr.nl/publications/publications/2021/11/16/navigating-and-anticipating-in-uncertain-times>
- ²⁰ <https://2.wetenschapsagenda.nl/route/op-weg-naar-veerkrachtige-samenlevingen/>
- ²¹ https://ec.europa.eu/health/funding/eu4health-2021-2027-vision-healthier-european-union_en
- ²² Louise Biddle, Katharina Wahedi, Kayvan Bozorgmehr, Health system resilience: a literature review of empirical research, *Health Policy and Planning*, Volume 35, Issue 8, October 2020, Pages 1084–1109, <https://doi.org/10.1093/heapol/czaa032>
- ²³ Topp, Stephanie M. 2020. "Power and politics: the case for linking resilience to health system governance." *BMJ Global Health*:e002891
- ²⁴ van de Bovenkamp, Hester M., Annemiek Stoopendaal, and Roland Bal. 2017. "Working with layers: The governance and regulation of healthcare quality in an institutionally layered system." *Public Policy and Administration* 32 (1):45-65. doi: 10.1177/0952076716652934.
- ²⁵ Elissen, A., Nolte, E., Hinrichs, S., Conklin, A., Adams, J., Cadier, B., ... & Vrijhoef, H. J. (2014). Evaluating chronic disease management in real-world settings in six European countries: Lessons from the collaborative DISMEVAL project. *International Journal of Care Coordination*, 17(1-2), 25-37.
- ²⁶ Seward, N., Hanlon, C., Hinrichs-Kraples, S., Lund, C., Murdoch, J., Salisbury, T. T., ... & Sevdalis, N. (2021). A guide to systems-level, participatory, theory-informed implementation research in global health. *BMJ global health*, 6(12), e005365.
- ²⁷ Gravesteyn B, Krijkamp E, Busschbach J, Geleijnse G, Helmrich IR, Bruinsma S, van Lint C, van Veen E, Steyerberg E, Verhoef K, van Saase J, Lingsma H, Baatenburg de Jong R; Value Based Operation Room Triage team collaborators. Minimizing Population Health Loss in Times of Scarce Surgical Capacity During the Coronavirus Disease 2019 Crisis and Beyond: A Modeling Study. *Value Health*. 2021 May;24(5):648-657.
- ²⁸ Scheduling surgery groups considering multiple downstream resources. A.J. Schneider, M. Carlier, J.T. van Essen, E.W. Hans. *European Journal of Operational Research* 282 (2): 741-752, 2020

²⁹ Weenink JW, Wallenburg I, Leistikow I, & Bal, R. Publication of inspection frameworks: a qualitative study exploring the impact on quality improvement and regulation in three healthcare settings *BMJ Quality & Safety* 2021;30:804-811.

³⁰ Leistikow, I., & Bal, R. (2020). Resilience and regulation, an odd couple? Consequences of Safety-II on governmental regulation of healthcare quality. *BMJ Quality & Safety*, 29, 869–872.

³¹ Van de Walle, B., Bruggemans, B. & Comes, T. (2016) Improving Situation Awareness in Crisis Response Teams: the Effect of Enriched and Centralized Coordination. *International Journal of Human Computer Studies*. 95, 66-79.

³² Chung, K. S. K.; Komashie, A.; and Yorke-Smith, N. Envisioning Complexity in Healthcare Systems using Discrete Event Simulation and Social Network Analysis. Proceedings of 5th IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM'13), Demonstration Track. Niagara Falls, Canada. August 2013.

³³ Kuntz, L., Scholtes, S., & Sülz, S. (2019). Separate and concentrate: Accounting for patient complexity in general hospitals. *Management Science*, 65(6), 2482-2501.

³⁴ Miedaner, F., & Sülz, S. (2020). Boundaries of focus and volume: An empirical study in neonatal intensive care. *Production and Operations Management*, 29(2), 298-308.

³⁵ Boulding, H., & Hinrichs-Krapels, S. (2021). Factors influencing procurement behaviour and decision-making: an exploratory qualitative study in a UK healthcare provider. *BMC health services research*, 21(1), 1-11.

³⁶ Hinrichs-Krapels, S., Ditewig, B., Boulding, H., Chalkidou, A., Erskine, J., & Shokraneh, F. (2021). Purchasing high-cost medical equipment in hospitals: A systematic review. *BMJ Open* (in review)

³⁷ Van Baal P, Perry-Duxbury M, Bakx P, Versteegh M, van Doorslaer E, Brouwer W. 2019. A cost-effectiveness threshold based on the marginal returns of cardiovascular hospital spending. *Health Economics*: 87-100;

³⁸ van Baal, P., Morton, A., & Severens, J. L. (2018). Health care input constraints and cost effectiveness analysis decision rules. *Social Science & Medicine*, 200, 59-64.

³⁹ Staňková, K., Brown, J.S., Dalton, W. and Gatenby, R. A., 2019. Optimizing cancer treatment using game theory. *JAMA Oncology* 5(1), 96-103.

⁴⁰ Boulding, H., & Hinrichs-Krapels, S. (2021). Factors influencing procurement behaviour and decision-making: an exploratory qualitative study in a UK healthcare provider. *BMC health services research*, 21(1), 1-11.

⁴¹ van Wyk, J. T., van Wijk, M. A., Sturkenboom, M. C., Mosseveld, M., Moorman, P. W., & van der Lei, J. (2008). Electronic alerts versus on-demand decision support to improve dyslipidemia treatment: a cluster randomized controlled trial. *Circulation*, 117(3), 371-378.

⁴² Lukosch, H. K., Bekebrede, G., Kurapati, S., & Lukosch, S. G. (2018). A scientific foundation of simulation games for the analysis and design of complex systems. *Simulation & Gaming*, 49(3), 279-314.

⁴³ P. Bakx. How to Talk About Healthcare Costs. Available at:
<https://www.youtube.com/watch?v=a5OINcJSqPA>

⁴⁴ OR triage decision tool. Gravesteijn, B. Accessed March 14 2022.
https://bgravesteijn.shinyapps.io/or_decision_making_shiny_app/

⁴⁵ Source: <https://www.ntvg.nl/gz>

⁴⁶ Hinrichs-Krapels, S., Bailey, J., Boulding, H., Duffy, B., Hesketh, R., Kinloch, E., ... & Grant, J. (2020). Using policy Labs as a process to bring evidence closer to public policymaking: a guide to one approach. *Palgrave Communications*, 6(1), 1-9.

⁴⁷ From good to better: new Dutch guidelines for economic evaluations in healthcare. *Pharmacoeconomics*, 34(11), 1071-1074.)

Date 8 March 2022
Contact person Prof.dr. M.E. Warnier
Telephone +31 (0)15 27 82232
Email M.E.Warnier@tudelft.nl
Subject Letter of Support
Ref 22001/MW/mt



Delft University of Technology

Faculty of Technology, Policy and
Management

Visiting address
Jaffalaan 5
2628 BA Delft
Correspondence
P.O. Box 5015
2600 GA Delft
The Netherlands

Dear Convergence Health & Technology Management Committee,

As Head of Department for the Multi-Actor Systems at the Faculty of Technology, Policy and Management of Delft University of Technology, I would like to express my support for the application 'Resilient HealthCare', led by Dr Saba Hinrichs-Krapels.

Our department is focussed on advancing knowledge and impact of socio-technical systems, with a particular focus on how different actors in such a system behave, are interdependent, and contribute to its outcomes. Examining the 'resilience' of healthcare is therefore a core topic for its future functioning and builds on our existing strengths in resilience of sociotechnical systems. We are particularly pleased that the Gamelab¹ facilities at our faculty are part of this proposal, which will enable the development of training materials for health practitioners and researchers, as well as provide opportunities to engage our bachelor and masters students in the continuous outputs from this programme, should it be awarded.

We particularly welcome the opportunity to work towards the mission of the Health and Technology branch of the Convergence agenda, thereby leveraging our collaborations with Erasmus MC and Erasmus University Rotterdam in multi-disciplinary work that can lead to societal impact.

Yours sincerely,

A handwritten signature in blue ink, appearing to read 'Martijn Warnier'.

Prof.dr. Martijn Warnier
Professor of Complex Systems Design
Head of the [Multi-Actor Systems Department](#)
[Delft University of Technology](#)

¹ <https://seriousgaming.tudelft.nl>

Budget formulier: Convergentie Flagship Call

Flagship:	Resilient Healthcare
Flagship acronym:	Resilient Healthcare
Naam Penvoerder:	Saba Hendriks-Kapels
Startdatum en looptijd van het project:	01-09-2022, 5 years
Totaal project budget:	€ 2.000.000,- by Convergence (€ 400.000,- per year) and € 2.000.000,- by matched funding

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	2022								2023		2024		2025		2026		2027		Total		Total per partner
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Trede	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoekorganisaties																					
	ErasmusMC	management assista	0.20	01/09/2022	31/08/2027	UMC - 8	10	€ 3,863	€ 69.534	€ 4.279	€ 13.015	€ 13.625	€ 14.159	€ 14.722	€ 9.734	€ 69.534	€ -	€		69.534	
	ErasmusMC	Medical PhD student	1.00	01/09/2022	31/08/2026	UMC - 10	7	€ 4.080	€ 293.760	€ 22.419	€ 69.302	€ 72.678	€ 76.702	€ 52.699	€ 293.760	€ -	€			293.760	
	ErasmusMC	ICT developer	1.00	01/09/2024	31/08/2027	UMC - 9	8	€ 4.370	€ 235.980	€ -	€ -	€ 24.570	€ 75.070	€ 80.381	€ 55.959	€ 235.980	€ -	€		235.980	
	ErasmusMC	postdoc	1.00	01/09/2024	31/08/2027	UMC - 10	12	€ 4.800	€ 259.200	€ 259.200	€ -	€ 27.473	€ 81.853	€ 87.364	€ 62.510	€ 259.200	€ -	€		259.200	
	TU Delft	OIO	1.00	01/09/2022	31/08/2026	UNI - p	3	€ 3.122	€ 224.784	€ 14.451	€ 49.298	€ 56.718	€ 60.514	€ 43.803	€ 43.803	€ 224.784	€ -	€		224.784	
	TU Delft	postdoc	1.00	01/09/2024	31/08/2027	UNI - 10	10	€ 4.222	€ 227.988	€ 227.988	€ -	€ 23.568	€ 72.078	€ 78.427	€ 55.915	€ 227.988	€ -	€		227.988	
	OIO	OIO	1.00	01/09/2022	31/08/2026	UNI - p	3	€ 3.122	€ 224.784	€ 14.451	€ 49.298	€ 56.718	€ 60.514	€ 43.803	€ 43.803	€ 224.784	€ -	€		224.784	
	EUR	OIO	1.00	01/09/2022	31/08/2026	UNI - p	3	€ 3.122	€ 224.784	€ 14.451	€ 49.298	€ 56.718	€ 60.514	€ 43.803	€ 43.803	€ 224.784	€ -	€		224.784	
Ondernemingen																					
	Onderneming 1							€ -	€ 0									€ 0	€ 0	€ 0.00	
	Onderneming 2							€ -	€ 0									€ 0	€ 0	€ 0.00	
	Onderneming 3							€ -	€ 0									€ 0	€ 0	€ 0.00	
	Onderneming 4							€ -	€ 0									€ 0	€ 0	€ 0.00	
Overige (private) partners																					
	Europese Unie - Horizon 2020 (awarded grant)	div personeel	1.00	01/10/2022	31/08/2024	UNI - 11	5	€ 4.490	€ 154.905	€ 30.000	€ 75.000	€ 49.905	€ -			€ 154.905	€ -	€		154.905	
	Europese Unie - Horizon 2020 (awarded grant)	div personeel	0.30	01/12/2022	31/08/2024	UNI - 11	6	€ 4.629	€ 43.744	€ 8.500	€ 21.250	€ 13.994	€ -			€ 43.744	€ -	€		43.744	
	Ministry of Health (awarded grant)	div personeel	2.00	01/09/2022	31/08/2024	UNI - 11	5	€ 4.490	€ 323.280	€ 64.000	€ 200.000	€ 99.280	€ -			€ 323.280	€ -	€		323.280	
	Healthcare Inspectorate (awarded grant)	div personeel	1.00	01/09/2022	31/08/2024	UNI - 10	10	€ 4.222	€ 151.992	€ 25.000	€ 75.000	€ 51.992	€ -	€ -	€ -	€ 151.992	€ -	€		151.992	
	EU European Training Network (awarded grant)	div personeel	0.20	01/09/2022	31/08/2024	UNI - 8	3	€ 3.122	€ 22.478	€ 3.722	€ 11.167	€ 7.589	€ -			€ 22.478	€ -	€		22.478	
	Europese Unie - Horizon 2020 (awarded grant)	div personeel	1.00	01/09/2022	31/12/2023	UNI - 11	9	€ 5.033	€ 120.750	€ 60.000	€ 60.750					€ 120.750	€ -	€		120.750	
	Dutch Ministry of Defense (awarded grant)	div personeel	1.25	01/09/2022	31/08/2024	UNI - 10	9	€ 4.093	€ 184.185	€ 24.000	€ 96.000	€ 64.185	€ -			€ 184.185	€ -	€		184.185	
	Centre for safety in healthcare (awarded grant)	div personeel	0.85	01/09/2022	31/08/2024	UNI - 10	3	€ 3.282	€ 100.429	€ 14.451	€ 49.298	€ 36.680	€ -			€ 100.429	€ -	€		100.429	
	Horizon Europe (pending future grant)	div personeel	3.15	01/09/2024	31/12/2026	UNI - 11	8	€ 4.629	€ 612.417	€ -	€ -	€ 102.608	€ 307.825		€ 201.984	€ 612.417	€ -	€		612.417	
	NWO VIDI (pending future grant)	div personeel	2.00	01/09/2024	31/08/2027	UNI - 10	10	€ 4.222	€ 455.976	€ -	€ -	€ 51.976	€ 170.000	€ 156.000	€ 78.000	€ 455.976	€ -	€		455.976	
	Naam 4							€ -	€ 0							€ -	€ -	€		-	
Totaal									€ 3,931,812											€ 3,931,812	
Totaal loonkosten																					
2.	2022								2023		2024		2025		2026		2027		Total		Total per partner
	Partner die de kosten maakt	Materiaal / hulpmiddel						Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoekorganisaties																					
	ErasmusMC	Materials & catering for interviews, workshops and trainings						€ 7.186	€ -		€ 500	€ 500	€ 500	€ 500	€ 3.000	€ 2.686	€ 7.186	€ -	€	7.186	
	Organisatie 2																€ -	€ -	€	-	
	Organisatie 3																€ -	€ -	€	-	
	Organisatie 4																€ -	€ -	€	-	
Ondernemingen																					
	Onderneming 1																	€ 0	€ 0	€ 0.00	
	Onderneming 2																	€ 0	€ 0	€ 0.00	
	Onderneming 3																	€ 0	€ 0	€ 0.00	
	Onderneming 4																	€ 0	€ 0	€ 0.00	
Overige (private) partners																					
	Naam 1																	€ 0	€ 0	€ 0.00	
	Naam 2																	€ 0	€ 0	€ 0.00	
	Naam 3																	€ 0	€ 0	€ 0.00	
	Naam 4																	€ 0	€ 0	€ 0.00	
Totaal									€ 7.186											€ 7.186	
Totaal kosten materialen en hulpmiddelen																					
3.	2022								2023		2024		2025		2026		2027		Total		Total per partner
	Partner die de kosten maakt	Machine / apparatuur						Prijs per gebruikseenheid	Gebbruik eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoekorganisaties																					
	Organisatie 1																	€ 0	€ 0	€ 0.00	
	Organisatie 2																	€ 0	€ 0	€ 0.00	
	Organisatie 3																	€ 0	€ 0	€ 0.00	
	Organisatie 4																	€ 0	€ 0	€ 0.00	
Ondernemingen																					
	Onderneming 1																	€ 0	€ 0	€ 0.00	
	Onderneming 2																	€ 0	€ 0	€ 0.00	
	Onderneming 3																	€ 0	€ 0	€ 0.00	
	Onderneming 4																	€ 0	€ 0	€ 0.00	
Overige (private) partners																					
	Naam 1																	€ 0	€ 0	€ 0.00	
	Naam 2																	€ 0	€ 0	€ 0.00	
	Naam 3																	€ 0	€ 0	€ 0.00	
	Naam 4																	€ 0	€ 0	€ 0.00	
Totaal									€ 0											€ 0	
Totaal kosten machines en apparatuur																					
4.	2022								2023		2024		2025		2026		2027		Total		Total per partner
	Partner die de kosten maakt	Naam derde, omschrijving kosten						Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoekorganisaties																					
	ErasmusMC	Statistics Netherlands (CBS), access to external data						€ 10.000			€ 10.000							€ 10.000	€ 0	€ 10.000.00	
	TU Delft	Subcontracting research assistants via FlexDelft for software development of serious games						€ 85.000					€ 50.000		€ 35.000			€ 85.000	€ 0	€	
	TU Delft	Game company (e.g. In There, The Bam, Raccoon Games), valuation of serious games						€ 100.000					€ 50.000		€ 50.000			€ 100.000	€ 0	€ 48.000.00	
	Organisatie 4																	€ 0	€ 0	€ 0.00	
Ondernemingen																					
	Onderneming 1																	€ 0	€ 0	€ 0.00	
	Onderneming 2																	€ 0	€ 0	€ 0.00	
	Onderneming 3																	€ 0	€ 0	€ 0.00	
	Onderneming 4																	€ 0	€ 0	€ 0.00	
Overige (private) partners																					
	Naam 1																	€ 0	€ 0	€ 0.00	
	Naam 2																	€ 0	€ 0	€ 0.00	
	Naam 3																	€ 0	€ 0	€ 0.00	
	Naam 4																	€ 0	€ 0	€ 0.00	
Totaal									€ 195.000											€ 195.000	
Totaal kosten derden																					
5.	2022								2023		2024		2025		2026		2027		Total		Total per partner
	Partner die de kosten maakt	Naam derde, omschrijving kosten						Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoekorganisaties																					
	ErasmusMC	Travel & conferences (€ 1.000,- per scientific FTE per year)						€ 7.000	€ 467		€ 1.400	€ 1.400.000	€ 1.400	€ 1.400	€ 1.400.000	€ 933.33	€ 7.000	€ 0		€ 7.000.00	
	TU Delft	Travel & conferences (€ 1.000,- per scientific FTE per year)						€ 7.000	€ 467		€ 1.400	€ 1.400.000	€ 1.400	€ 1.400	€ 1.400.000	€ 933.33	€ 7.000	€ 0		€ 7.000.00	
	EUR	Travel & conferences (€ 1.000,- per scientific FTE per year)						€ 8.000	€ 533		€ 1.800	€ 1.800.000	€ 1.800	€ 1.800	€ 1.800.000	€ 1.066.67	€ 8.000	€ 0		€ 8.000.00	
	ErasmusMC	Open access publications						€ 7.500			€ 1.500	€ 1.500	€ 2.000	€ 2.000	€ 2.500	€ 7.500	€ 0			€ 7.500.00	
	EUR	Open access publications						€ 7.500			€ 1.500	€ 1.500	€ 2.000	€ 2.000	€ 2.500	€ 7.500	€ 0			€ 7.500.00	
Ondernemingen																					
	Onderneming 1																	€ 0	€ 0	€ 0.00	
	Onderneming 2																	€ 0	€ 0	€ 0.00	
	Onderneming 3																	€ 0	€ 0	€ 0.00	
	Onderneming 4																	€ 0	€ 0	€ 0.00	
Overige (private) partners																					
	Naam 1																	€ 0	€ 0	€ 0.00	
	Naam 2																	€ 0	€ 0	€ 0.00	
	Naam 3																	€ 0	€ 0	€ 0.00	
	Naam 4																	€ 0	€ 0	€ 0.00	
Totaal									€ 37.800											€ 37.800	
Totaal kosten derden																					
6.	2022								2023		2024		2025		2026		2027		Total		Total per partner
	Partner die de kosten maakt																				

FINANCIERING - Bijdragen per partner per jaar

in Euro		2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
1	Organisatie 7	€ 0		€ 0		€ 0		€ 0		€ 0		€ 0		€ 0	€ 0	€ 0
2	Organisatie 7	€ 0		€ 0		€ 0		€ 0		€ 0		€ 0		€ 0	€ 0	€ 0
3	Organisatie 8	€ 0		€ 0		€ 0		€ 0		€ 0		€ 0		€ 0	€ 0	€ 0
...	Organisatie 4													€ 0	€ 0	€ 0
	Totale bijdrage onderzoeksorganisaties	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Ondernemingen																
4	Ondememing 1													€ 0	€ 0	€ 0
5	Ondememing 2													€ 0	€ 0	€ 0
6	Ondememing 3													€ 0	€ 0	€ 0
...	Ondememing 4													€ 0	€ 0	€ 0
	Totale bijdrage ondernemingen	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Overige (private) partners																
7	Europese Unie - Horizon 2020 (awarded grant)	€ 98,500		€ 157,042		€ 63,899		€ 0		€ 0		€ 0		€ 319,441	€ 0	€ 319,441
8	Ministry of Health (awarded grant)	€ 64,000		€ 200,000		€ 59,280		€ 0		€ 0		€ 0		€ 323,280	€ 0	€ 323,280
9	Healthcare inspectorate (awarded grant)	€ 25,000		€ 75,000		€ 51,992		€ 0		€ 0		€ 0		€ 151,992	€ 0	€ 151,992
	EU European Training Network (awarded grant)	€ 3,722		€ 11,167		€ 7,589		€ 0		€ 0		€ 0		€ 22,478	€ 0	€ 22,478
	Dutch Ministry of Defense (awarded grant)	€ 24,000		€ 96,000		€ 64,185								€ 184,185	€ 0	€ 184,185
	Centre for safety in healthcare (awarded grant)	€ 14,451		€ 49,298		€ 36,680								€ 100,429	€ 0	€ 100,429
	Horizon Europe (pending future grant)					€ 102,608		€ 307,825		€ 201,984				€ 612,417	€ 0	€ 612,417
	NWO VIDI (pending future grant)					€ 51,976		€ 170,000		€ 156,000		€ 78,000		€ 455,976	€ 0	€ 455,976
...	Flagship	€ 133,333		€ 400,000		€ 400,000		€ 400,000		€ 400,000		€ 266,667		€ 2,000,000	€ 0	€ 2,000,000
	Totale overige bijdrage	€ 363,006	€ 0	€ 988,507	€ 0	€ 838,209	€ 0	€ 877,825	€ 0	€ 757,984	€ 0	€ 344,667	€ 0	€ 4,170,198	€ 0	€ 4,170,198
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Totaal budget per jaar		€ 363,006	€ 0	€ 988,507	€ 0	€ 838,209	€ 0	€ 877,825	€ 0	€ 757,984	€ 0	€ 344,667	€ 0	Totaal		
		€ 4,170,198														